

Clustering polycystic ovary syndrome laboratory results extracted from a large internet forum with machine learning

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ABSTRACT

Background: Polycystic Ovary Syndrome (PCOS) is reported to affect between 4% and 21% of reproductive aged people with ovaries. It is a heterogeneous condition with a lack of established phenotypes that address the range of reproductive and metabolic features present in PCOS. These reproductive and metabolic features may result in patients undergoing a variety of relevant laboratory tests. Previous work has led to the gathering of laboratory test results from a PCOS specific forum, hosted on a website called reddit.

Objectives: In this paper, laboratory results and body mass index (BMI) posted on the PCOS reddit forum were clustered to show the usefulness of the PCOS forum for PCOS research and validate existing PCOS phenotypes or discover other appropriate phenotypes.

Methods and results: Over 1500 sets of PCOS-related reddit laboratory test results and BMIs were clustered using nearest neighbour imputation and K-means clustering. However, only non-imputed data was included in the final clusters. Kernel Density Estimation plots were used to display the distinct clusters. The clustered test results suggested the existence of distinct metabolic and reproductive phenotypes, as well as a group displaying mild features of both types of dysregulations and a group skewed towards normal results. It was also possible to separate the groups further into distinct hypothyroid groups within the mixed dysregulation group and to separate insulin resistant and diabetes-like groups within the metabolic group.

Conclusions: This research further validates the usefulness of exploring alternate data sources in the age of the internet and machine learning. The reddit clusters reinforced the existing notion that people with PCOS can be separated into a primarily metabolic pathology group, a primarily reproductive pathology group and an in-between group with pathology in both domains.

1. Introduction

Polycystic Ovary Syndrome (PCOS) is a heterogeneous condition likely affecting between 4% and 21% of reproductive aged people with ovaries [1]. It is closely associated with hyperandrogenism [2,3], insulin resistance [4,5], infertility [6], obesity [7], type 2 diabetes (T2DM) [8,9], and hypothyroidism [10]. Diagnosis of PCOS currently requires at least two of the Rotterdam criteria (Table 1) and the exclusion of conditions that can mimic PCOS [11]. These exclusions could be hyperprolactinaemia, thyroid disease, non-classic congenital adrenal

hyperplasia, adrenal tumours and Cushing's syndrome [12].

The nature of the diagnosis leads to four commonly discussed phenotypes, with the first being a combination of all Rotterdam criteria. The remaining phenotypes are the three possible combinations of two Rotterdam criteria. However, these phenotypes do not consider the metabolic dysfunction that commonly affects people with PCOS. In order to separate those who do not suffer the metabolic dysregulation, some experts classify a subset of people with PCOS as having 'lean' PCOS [4,13]. However, insulin resistance is still pervasive in approximately 75% of 'lean' people with PCOS [4]. Other experts group PCOS based on

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Table 1
Rotterdam diagnostic criteria [11].

Hyperandrogenism	Clinical:	Hirsutism, acne and/or alopecia
	Biochemical:	Elevated free testosterone, total testosterone, free androgen index, androstenedione, and/or DHEA-S
Irregular cycles and ovulatory dysfunction	Irregular cycles, anovulation, oligo-ovulation	
Polycystic ovarian morphology	Over 20 follicles on either ovary An ovarian volume over 10 mL on either ovary The presence of large cysts or dominant follicles	

whether people present with metabolic and/or reproductive issues [14, 15]. Dapas et al. [14] supported their findings by clustering biochemical and genotype data, thereby providing biologically relevant phenotypes with their own genetic distinctions. However, to the authors' knowledge, there are no firmly established phenotypes that are clinically relevant to all the different presentations of PCOS. Much of the current research regarding clinically relevant phenotypes involves linking relevant factors such as metabolic abnormalities to the existing Rotterdam phenotypes [16]. However, many of the major decisions regarding PCOS diagnosis, and by extension the phenotypes of PCOS, were based on expert opinion rather than evidence [16]. The current lack of evidence-based phenotypes may be due to the scarcity of large datasets that could provide statistical inference for defining phenotypes.

Advances in machine learning and the prevalence of specific internet support groups have made it possible to process large amounts of data from freely available online sources. Such data includes large volumes of laboratory test results from people with clinically confirmed PCOS or those who suspect they may have PCOS. Clustering, a form of unsupervised machine learning, has also become increasingly popular with clinicians due to its success clarifying phenotypes that human approaches may have failed to detect [17]. In this paper, laboratory test results and BMIs from an online PCOS forum have been clustered to explore possible phenotypes within PCOS and further validate internet forums in clinical research. The clusters identified were compared to the clusters obtained by Dapas et al. [14].

2. Methods

2.1. Dataset

The dataset used in this study is the PCOS subreddit laboratory test result dataset gathered and processed in previous work [18]. Reddit users are invited to contribute to the forum in 'posts'. Generally these 'posts' lead to discussions or comments from other users in a 'comments' section of the post. In this research, all ~45,000 posts and ~300,000 comments from the PCOS subreddit [19] were downloaded in May 2021, via the Pushshift reddit dataset [20]. Posts likely to contain laboratory test results were selected by machine learning. These posts were read and any laboratory test results were manually recorded. Other potentially relevant results, such as age and BMI were also recorded when available. Sometimes recording results involved context-based estimation of the type of test and the unit. Any results with potentially poor estimation were marked as uncertain. Due to the noisy nature of the dataset, extreme outliers with an absolute z-score greater than 4 were also removed. After this pre-processing the dataset contained 1585 test sets. A full description of the data extraction and pre-processing steps is provided in Ref. [18].

The results reported for each user was dependent on what tests their health care provider ordered and which tests they chose to post. Thus, the final dataset contained large portions of 'missing values' when considering all possible tests within the dataset. It is important to note that these gaps in the data were not truly missing values. While the original dataset included age results, they were excluded from the

clustering as every user was in the age range associated with ovulation and age is not a typical factor in PCOS clinical phenotyping outside of adolescence. Including unnecessary but ranged results such as age would have added noise to the clustering procedure. Table 2 is a summary of the dataset, listing how often each test was present with respect to the total number of test sets. After only including test sets with the relevant results listed in Table 2, there were 1496 test sets left to cluster.

Table 2
Summary of the 1585 PCOS subreddit test sets used in the clustering.

Test Result	Mean ± SD	Unit Used	Number of Results reported	% of Test Sets with Result
Total Testosterone	55.9 ± 37.9	ng/dL	552	34.8
Dehydroepiandrosterone- Sulphate (DHEA-S)	396.5 ± 165.2	mcg/dL	381	24.0
Body Mass Index (BMI) at Time of Blood Test	24.4 ± 6.4	kg/m ²	240	15.1
Ratio of Luteinizing Hormone to Follicle Stimulating Hormone (LH/FSH)	2.2 ± 1.4	Unitless	236	14.9
Free Testosterone	5.9 ± 4.4	pg/mL	226	14.3
Fasting Plasma Glucose	94.2 ± 17.6	mg/dL	219	13.8
Follicle Stimulating Hormone (FSH)	5.8 ± 3.1	U/L	214	13.5
Glycated haemoglobin (HbA1c)	5.5 ± 0.8	%	211	13.3
Luteinizing Hormone (LH)	12.9 ± 13.6	U/L	206	13.0
Thyroid Stimulating Hormone (TSH)	2.4 ± 1.6	mU/L	176	11.1
Prolactin	24.2 ± 23.9	ng/mL	140	8.8
Oestradiol	79.3 ± 88.2	pg/mL	138	8.7
Fasting Insulin	12.9 ± 12.0	mU/L	131	8.3
Progesterone	5.0 ± 10.1	ng/mL	89	5.6
Sex Hormone Binding Globulin (SHBG)	84.4 ± 70.7	nmol/L	72	4.5
Anti-Müllerian Hormone (AMH)	10.2 ± 6.0	ng/mL	61	3.8
Homeostasis Model Assessment- Estimated Insulin Resistance (HOMA-IR) Index (Calculated using fasting insulin and glucose)	2.1 ± 2.4	–	57	3.6
Free Thyroxine	1.1 ± 0.2	ng/dL	48	3.0
17-Hydroxyprogesterone	125.7 ± 111.8	ng/dL	44	2.8
Vitamin D	26.4 ± 17.3	ng/mL	41	2.6
Ferritin	37.8 ± 33.0	ng/mL	31	2.0
Total Cholesterol	219.8 ± 55.2	mg/dL	30	1.9
Low-Density Lipoprotein Cholesterol	132.1 ± 46.6	mg/dL	27	1.7
Triglycerides	129.9 ± 75.7	mg/dL	25	1.6
Androstenedione	349.9 ± 376.3	ng/dL	23	1.5
Alanine Transaminase	90.0 ± 106.4	U/L	23	1.5
High-Density Lipoprotein Cholesterol	60.7 ± 18.8	mg/dL	23	1.5

2.2. Clustering

The incomplete nature of the subreddit dataset meant that clustering could not be performed on the raw data. A common way to rectify gaps in a dataset is to impute the missing data before applying techniques such as clustering [21]. However, the subreddit dataset had significantly large proportions of data that needed imputing. Therefore, imputation was treated as a pre-processing step in obtaining the clusters and assigning each test set to a cluster. Only non-imputed data from each test set is included in the final clusters.

In this research, gaps in the data were estimated using the KNN (k-Nearest Neighbour) imputer from the SciKit-Learn python toolkit [22]. Nearest neighbour imputation is a technique where the closest neighbours to a datum are used to estimate the missing values in the datum. In a large survey of imputation methods, KNN imputation was shown to yield competitive results when compared to both simple and complex imputation methods [21]. For the KNN imputation in this paper, gaps in the data were estimated by averaging the 7 nearest neighbours according to Euclidean distance. This is described in more detail within Appendix A.

Due to the large variations between the different test results, some form of scaling was also necessary before clustering could occur. The SciKit-Learn standard scalar was applied after imputation, which scaled data to its corresponding z-score, giving (x_{ji}) as the standardised score for test j in test set i .

Currently, there are a number of distinct clustering approaches in the literature, with each exhibiting beneficial behaviour in certain datasets [23]. K-mean clustering using the SciKit-Learn toolkit was selected for this research as K-mean clustering is widely used in clinical dataset analysis [24]. The clustering algorithm is shown in Fig. 1, where clustering is complete when the within-cluster sum of squares (WSS) converges to a minimum. Equation (1) shows the WSS criterion where ' n ' is the number of points in a specific test, ' C ' is the set of cluster centroids, ' k ' is the cluster number out of the total number of clusters (K), ' x_i ' is the vector of all real and imputed results for a test set and ' μ_k ' is the centroid

for a specific cluster. Convergence is achieved when the difference between the old centroids and the new centroids is less than a certain threshold. The threshold used was 0.0001 and the convergence criteria is shown in Equation (2), where ' th ' is the threshold and ' μ_0 ' represents the centroids from the previous iteration. The SciKit-Learn Kmeans function also performs the algorithm a default of ten times with different initial centroids and returns the clusters with the smallest WSS criterion.

$$WSS_criterion = \sum_{i=1}^n \min_{\mu_k \in C} (\|x_i - \mu_k\|^2) \quad \text{Eq. 1}$$

$$\sum_{k=1}^K (\mu_{0k} - \mu_k)^2 < th \quad \text{Eq. 2}$$

2.3. Validation

A variety of techniques were used to undertake internal and external validation of the clusters. Clustering was performed on the entire dataset, then to assess stability clustering was repeated on a randomly selected 90% of the dataset. After clustering of 90% of the dataset was completed, the remaining 10% of the data was added to the closest cluster by comparing the Euclidean distance of the cluster means with each datum to be added. The clusters made by clustering 90% of the dataset and adding 10% afterwards will be referred to as the clusters from the 'split dataset'. The clusters from the complete dataset and the split dataset were visually compared using kernel density estimation (KDE) plots to validate the stability of the clusters.

The elbow method was used to help select the appropriate number of clusters. To use the elbow method, clustering was undertaken with $K \in [2, 8]$. The fit score (Equation (3)) was calculated for each K using standard deviations (σ). Plots of K vs fit_score were made for both the full and split dataset. The elbow method uses an ad-hoc interpretation of the graphs to imply the appropriate number of clusters. In particular, as the number of clusters grows, there is typically a saturation in the improvement of fit_scores . Coarsely described, the optimal number of

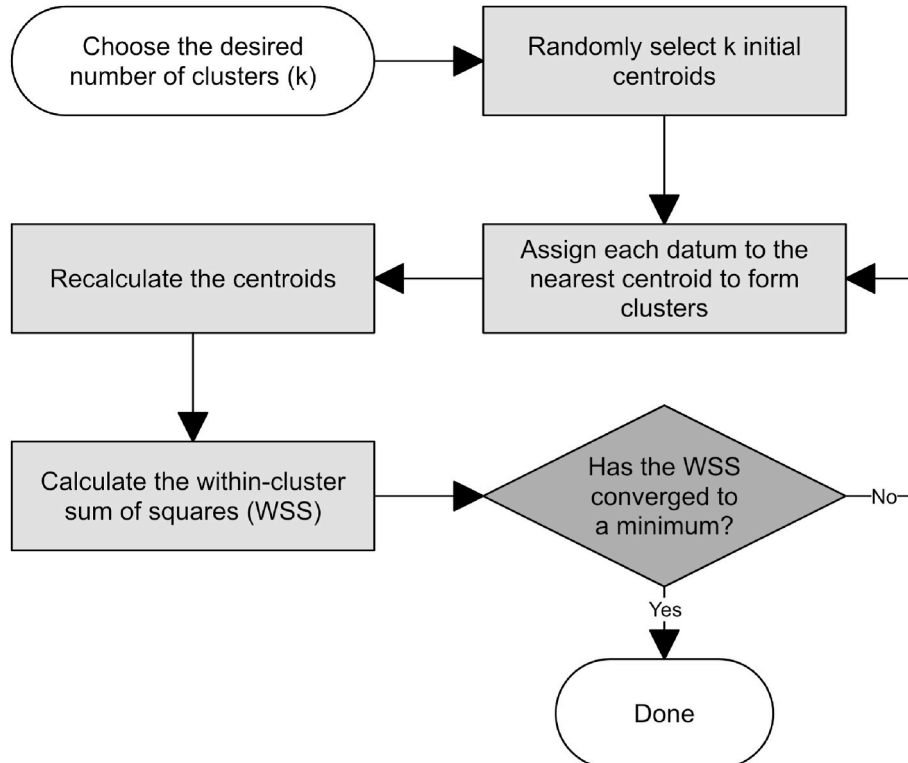


Fig. 1. Diagram of the K-means clustering using in this paper.

clusters is typically declared at an ‘elbow’, or bend, in the graph.

$$fit_score_K = \frac{\sum_{k=1}^K \sum_{j=1}^J \sigma_{kj}}{\sum_{j=1}^J \sigma_j} \quad \text{Eq. 3}$$

If the elbow plots indicated a cluster number was of interest, the clusters of that number were explored further. KDE was used to plot both groups of clusters with respect to one type of laboratory test. Only clusters with more than 10 points of the specific test were included in KDE plots. Only KDE plots with at least two included clusters and one cluster with both all and split data included were considered. These plots were used to assess which clusters from the split data corresponded to the clusters from all the data.

The final method of validation involved coarsely comparing the subreddit results to the published results from the work of Dapas et al. [14], who also clustered PCOS data. They found a reproductive group with higher LH and SHBG levels as well as lower BMI and fasting insulin (23%), a metabolic group with higher BMI, fasting glucose and fasting insulin as well as lower SHBG and LH levels (37%), and labelled the remaining data as indeterminate (40%) [14].

3. Results

3.1. Choosing clusters

The elbow plots of the clusters found in all the data and the split data, when clustering was performed on 90% of the data and the remaining 10% was added post-clustering, are shown in Fig. 2. They both had a consistent bend at four clusters and thus the four cluster groupings were explored in the next section. The plot made from all the data has another bend at seven clusters and the split data plot has another bend at six clusters. Thus, either six or seven clusters could have been of interest. Seven clusters were chosen for further analysis along with the more definitive four clusters as the combined fit scores for seven clusters were lowest.

3.2. Four clusters

Four clusters were chosen as the most stable groups across both the complete and the split datasets. Table 3 shows the number of discrete results in each cluster and a summary of the cluster features. The four cluster KDE plots for the androgens, reproductive hormones, and metabolic tests are shown in Figs. 3–5, respectively. While z-scores were used during clustering, the scale of the x-axes are dependent on absolute values to improve understanding. The reference ranges (or reference intervals) for each test are shown with grey shading and the references

Table 3

Number of test results and features of each cluster for four clusters.

	Cluster 1 ₄		Cluster 2 ₄		Cluster 3 ₄		Cluster 4 ₄	
	All	Split	All	Split	All	Split	All	Split
Discrete Results	203	250	649	607	447	448	197	191
Testosterone	HN	H	N	N	H	H	HN	HN
DHEA-S	H	H	N	N	H	H	–	HN
LH/FSH	–	–	N	N	H	H	–	N
Prolactin	–	H	N	N	HN	HN	–	–
AMH	VH	–	HN	HN	–	H	HN	–
BMI	N	HN	N	N	HN	HN	H	H
Fasting Glucose	–	HN	N	N	N	N	H	H
HbA1c	–	–	N	N	N	N	HN	HN
TSH	–	H	N	N	H	H	N	N
Fasting Insulin	–	–	N	N	N	N	H	H
Overall Label	Mixed/ Undefined		Normal		Reproductive		Metabolic	

VH, very high; H, high; HN, high normal; N, normal. DHEA-S, dehydroepiandrosterone-sulphate; LH/FSH, the ratio of luteinising hormone to follicle stimulating hormone; AMH, anti-Müllerian hormone; BMI, body mass index; HbA1c, glycated haemoglobin; TSH, thyroid stimulating hormone.

for these ranges are given in Appendix B. The number of test results for each test and cluster are given in the figure legends. Imputed values were omitted from the graphs as they were used solely for cluster optimisation, and thus do not contribute to cluster strength. Hence, due to the sparsity of the data, there are some clusters that do not contain information about certain tests.

3.3. Seven clusters

Seven clusters did not appear to be as stable as four clusters. The results are still potentially of interest, however, there is a lack of PCOS clustering studies in the literature that explore more than three groups. Thus, the seven clusters found in this study were difficult to externally validate. Table 4 gives overall cluster size and feature information. KDE plots for the seven cluster analysis are in Appendix C.

4. Discussion

4.1. Four clusters

Several interesting clusters were identified within the PCOS subreddit laboratory test dataset. The whole and split data clustering method showed the consistency and stability of several clusters. In particular, the four cluster results were very stable across the complete and split datasets (Figs. 3–5). Cluster 1₄ was characterised by higher androgens (Fig. 3) and higher total cholesterol (Fig. 5h). Features not

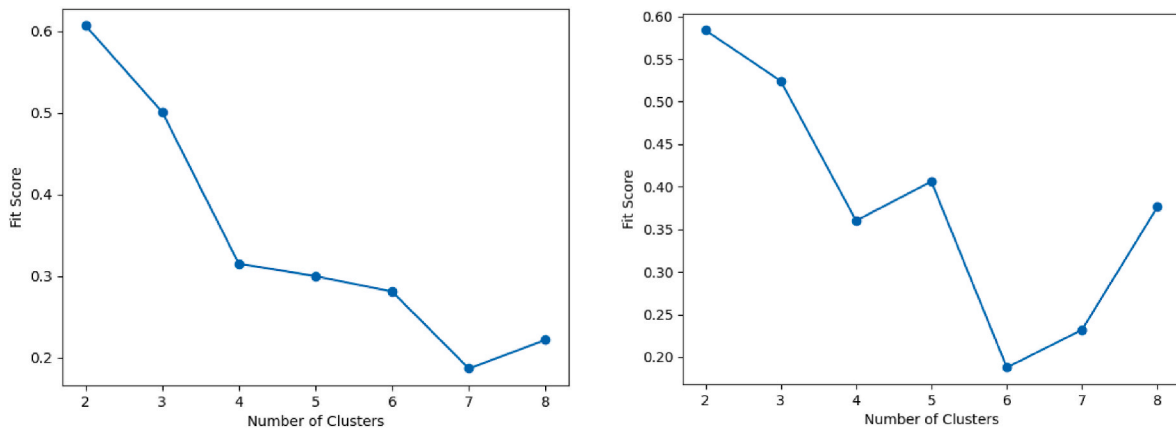


Fig. 2. Elbow plots of clusters within all the data (left) and the split data (right).

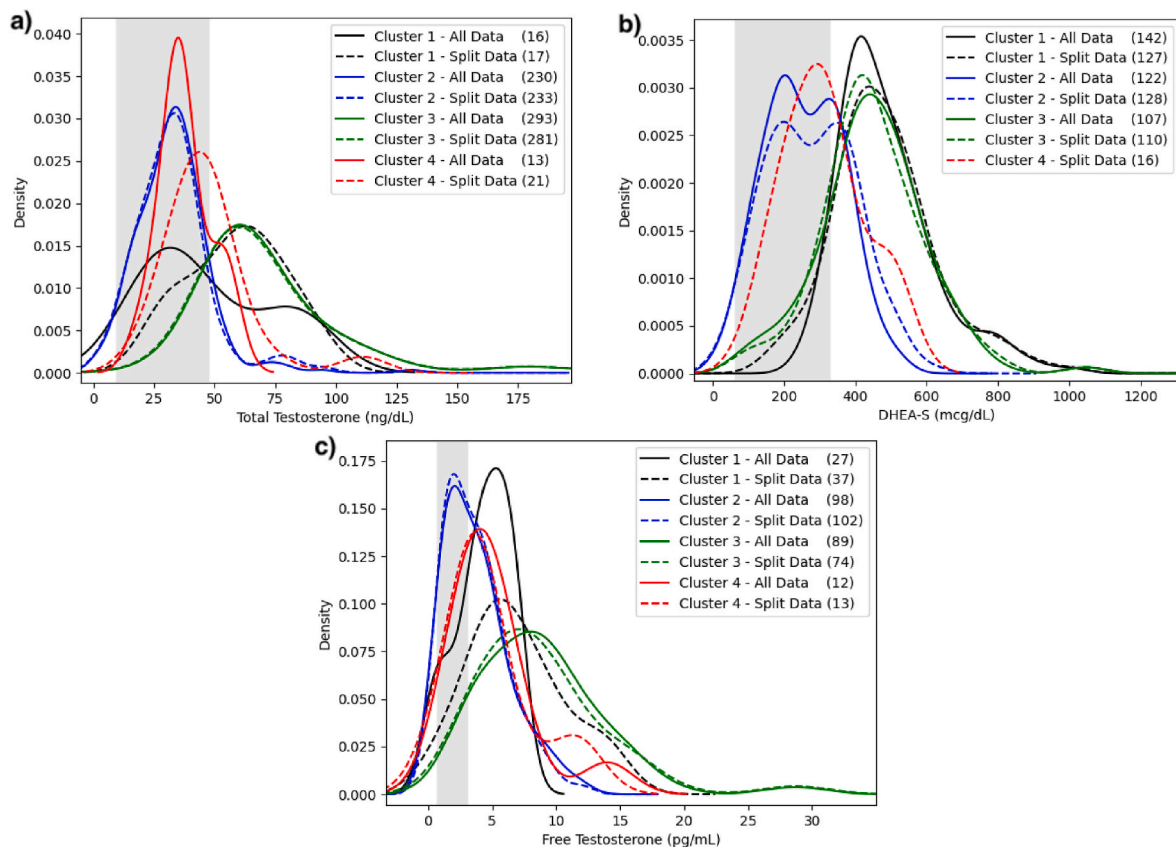


Fig. 3. Kernel density estimation plots showing the distribution of androgen data across four clusters for **a)** total testosterone, **b)** DHEA-S (dehydroepiandrosterone-sulphate) and **c)** free testosterone. Solid lines are the clusters made using the entire dataset. Dashed lines are the clusters made when the dataset was split. Reference ranges are shaded grey. Groups with fewer than ten data entries are not shown. Numbers in parentheses show the number of test results for a particular cluster.

validated by the stability analysis, but still potentially relevant, were high prolactin (Fig. 4d), high AMH (Fig. 4g), and moderately high BMI (Fig. 5a), fasting glucose (Fig. 5b) and TSH (Fig. 5d). As such, Cluster 1₄ displayed both reproductive and metabolic PCOS features. Cluster 2₄, the largest group (Table 3), had approximately normal levels for androgens (Fig. 3), reproductive hormones (Fig. 4) and metabolic tests (Fig. 5). Due to the self-reported and unselected nature of the dataset, it was expected that a portion of the users would not have clinically confirmed PCOS, and thus may be represented by Cluster 2₄. It is also possible for people to be diagnosed with PCOS based solely on clinical features such as history and examination, rather than biochemical evidence. Therefore, a group without PCOS features was expected.

Like Cluster 1₄, Cluster 4₄ also displayed high androgens (Fig. 3), however it lacked the higher fasting glucose (Fig. 5b), prolactin (Fig. 4d) and total cholesterol (Fig. 5h) of Cluster 1₄. Cluster 3₄ also skewed highest for LH/FSH (Fig. 4a) and TSH (Fig. 5d). Overall Cluster 3₄ appeared to be dominated by the reproductive pathology of PCOS while also including cases of dysfunctional thyroids. Cluster 4₄ was characterised by normal androgens (Fig. 3) and more normally skewed reproductive hormones (Fig. 4) but the most severe metabolic dysregulation (Fig. 5). Cluster 4₄ skewed highest for BMI (Fig. 5a), fasting glucose (Fig. 5b), HbA1c (Fig. 5c) and fasting insulin (Fig. 5e). Therefore, Cluster 4₄ appeared to be dominated by the metabolic side of PCOS and included people with diabetes. It is interesting to note that higher TSH seemed more associated with increased androgens than with metabolic dysregulation. A study of 800 women also found a link between hyperandrogenism and high TSH in PCOS [25].

Dapas et al. [14] clustered biochemical PCOS data with genotype data and also found a reproductive group (23 %), metabolic group (37 %) and an indeterminate group (40 %). The four clusters found in this paper contained a distinct reproductive group, a distinct metabolic

group, and another group containing both metabolic and reproductive features just as Dapas et al. [14] identified. However, our research also yielded a cluster of seemingly normal results and different proportions of individuals within the PCOS groups. This is likely due to differences in data sources. The subreddit data set, populated by self-selected contributors, may contain data that is clinically more heterogeneous when compared to that of Dapas et al. [14], who limited their data to individuals diagnosed by NIH diagnostic criteria. Other researchers have also made attempts to identify PCOS phenotypes through clustering [26, 27]. In van der Ham et al. [27], clustering of 2502 people fitting the Rotterdam criteria also revealed a reproductive, metabolic and indeterminate group proportioned similarly to the groups in Dapas et al. [14]. Elsayed et al. [26] clustered two smaller PCOS cohorts of 210 and 310 people. In both cohorts groups that could be labelled a reproductive, metabolic and indeterminate group were identified, though proportions differed between cohorts [26]. The group in Elsayed et al. [26] characterised by the highest testosterone and LH was similar to this study's Cluster 3₄, the group characterised by the highest BMI and HOMA-IR was similar to Cluster 4₄, and the group with the highest DHEA-S could correspond to Cluster 1₄.

Overall, the clusters presented in Table 3 support the notion that 'reproductive PCOS' and 'metabolic PCOS' are underpinned by distinct irregularities in endocrine function [14,15,26,27]. The agreement between the subreddit clusters found in this paper and the clusters found in the literature serve to confirm that there is a valid, identifiable PCOS population within the PCOS subreddit. This supports the notion that further research into the PCOS subreddit population could lead to useful conclusions about PCOS.

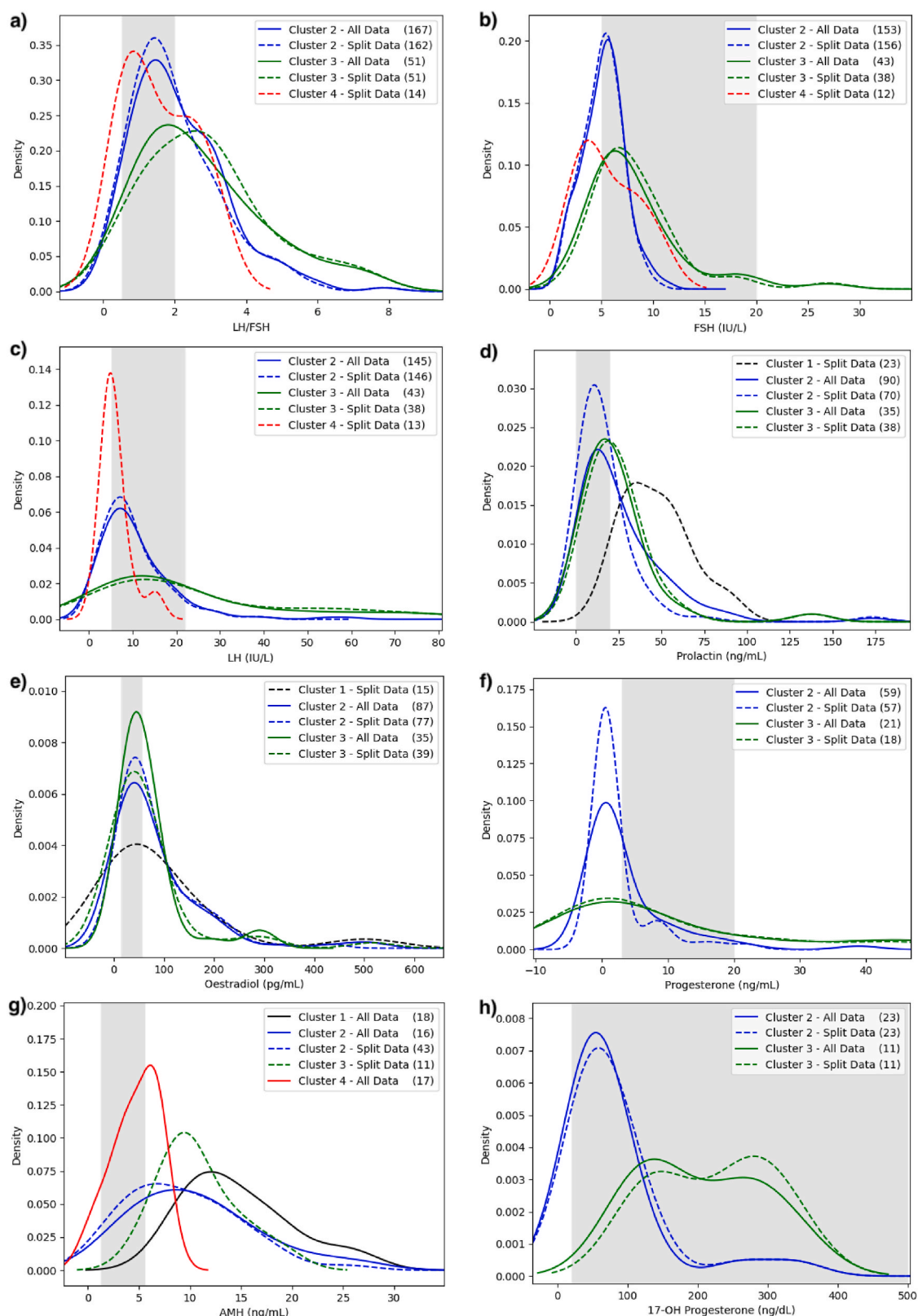


Fig. 4. Kernel density estimation plots showing the distribution of reproductive hormone data across four clusters for **a)** LH/FSH (the ratio of luteinising hormone to follicle stimulating hormone), **b)** FSH (follicle stimulating hormone), **c)** LH (luteinising hormone), **d)** prolactin, **e)** oestradiol, **f)** progesterone, **g)** AMH (anti-Müllerian hormone) and **h)** 17-Hydroxyprogesterone. Solid lines are the clusters made using the entire dataset. Dashed lines are the clusters made when the dataset was split. Reference ranges are shaded grey. Groups with fewer than ten data entries are not shown. Numbers in parentheses show the number of test results for a particular cluster.

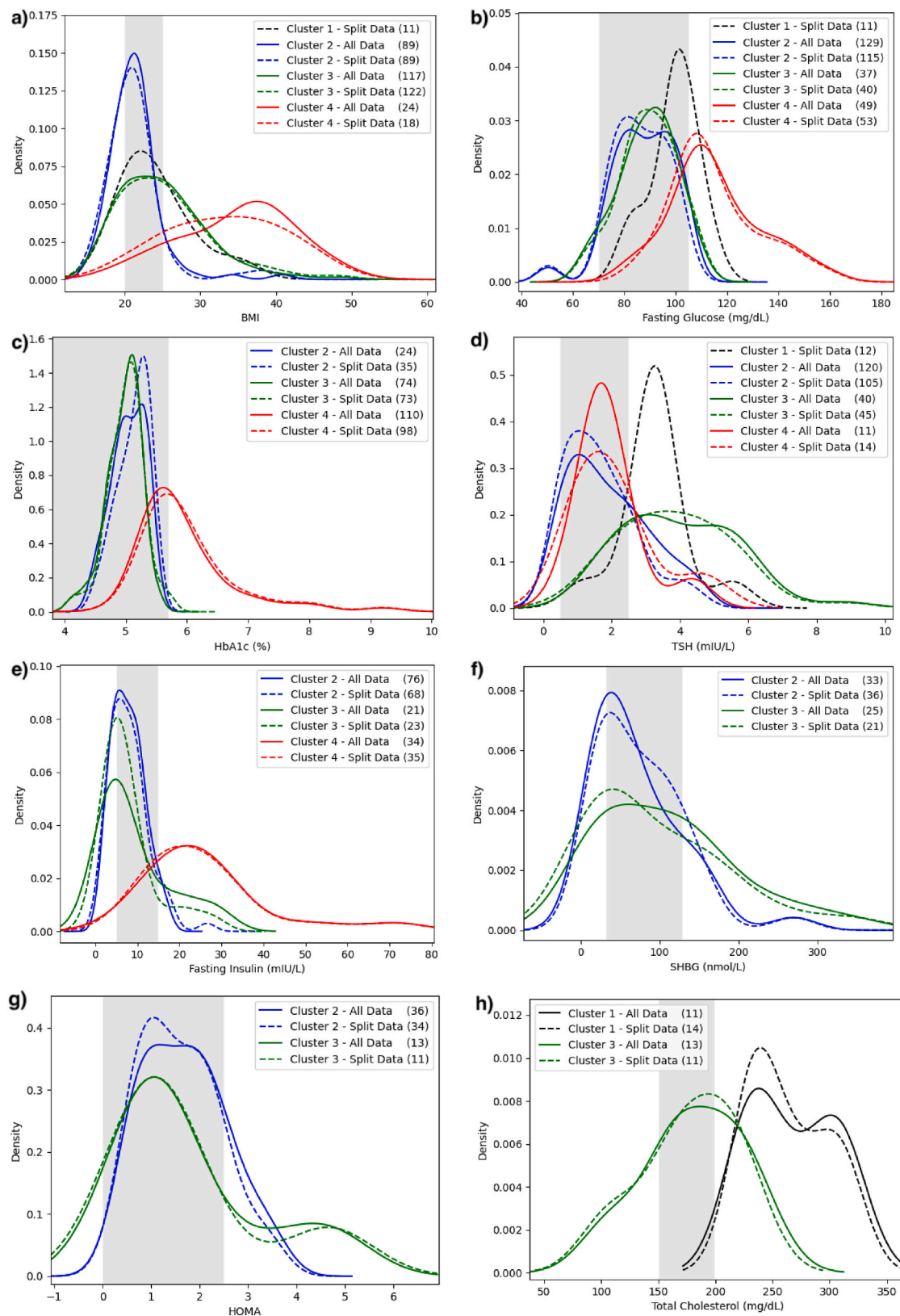


Fig. 5. Kernel density estimation plots showing the distribution of metabolic test data across four clusters for **a)** BMI (body mass index), **b)** fasting glucose, **c)** HbA1c (glycated haemoglobin), **d)** TSH (thyroid stimulating hormone), **e)** fasting insulin, **f)** SHBG (sex hormone binding globulin), **g)** HOMA-IR (homeostasis model assessment-estimated insulin resistance) and **h)** total cholesterol. Solid lines are the clusters made using the entire dataset. Dashed lines are the clusters made when the dataset was split. Reference ranges are shaded grey. Groups with fewer than ten data entries are not shown. Numbers in parentheses show the number of test results for a particular cluster.

Table 4

Number of test results and features of each cluster for seven clusters.

Test	Cluster 1 ₇		Cluster 2 ₇		Cluster 3 ₇		Cluster 4 ₇		Cluster 5 ₇		Cluster 6 ₇		Cluster 7 ₇	
	All	Split	All	Split	All	Split	All	Split	All	Split	All	Split	All	Split
Discrete Results	211	214	167	173	239	94	152	325	499	397	134	189	94	104
Testosterone	H	H	H	H	H	H	VH	VH	N	N	–	–	–	–
DHEA-S	H	H	N	HN	HN	N	H	H	N	N	–	VH	–	–
LH/FSH	N	HN	H	H	–	–	N	N	N	N	–	–	–	N
Prolactin	NH	H	H	H	HN	–	HN	HN	N	N	–	–	–	–
BMI	N	–	L	L	HN	HN	N	HN	N	N	–	–	H	H
Fasting Glucose	–	–	N	N	N	N	–	N	N	N	HN	N	–	H
HbA1c	–	–	N	–	N	N	–	N	N	N	–	–	HN	HN
TSH	N	–	–	N	H	–	HN	H	N	N	–	–	–	N
Fasting Insulin	–	–	–	–	N	–	–	N	N	N	H	H	–	–
Overall Label	Hyperandrogenism, hyperprolactinemia		High Testosterone, Reproductive Dysregulation, Underweight		High Testosterone, Overweight, Hypothyroidism		Severe Hyperandrogenism, Hypothyroidism		Normal		Insulin Resistant		Obese Prediabetic	

VH, very high; H, high; HN, high normal; N, normal; L, low. DHEA-S, dehydroepiandrosterone-sulphate; LH/FSH, the ratio of luteinising hormone to follicle stimulating hormone; AMH, anti-Müllerian hormone; BMI, body mass index; HbA1c, glycated haemoglobin; TSH, thyroid stimulating hormone.

4.2. Seven clusters

The seven cluster groupings (Appendix C) were less stable than the four cluster groupings. However, this was an expected outcome due to the lower statistical strength of groups with lower sample sizes. Nonetheless, the results could still be of interest because there may be subgroups beyond the initial four clusters that are clinically relevant to diagnosis and treatment of PCOS.

The seven clusters show several interesting trends that could not be captured within the four clusters (Appendix C). First, it is interesting that there is both an insulin resistant and a separate group with glucose and HbA1c levels that indicated diabetes (Clusters 6₇ and 7₇, respectively). Second, Cluster 3₇ seemed to be a hypothyroid group without any severe PCOS dysregulation or insulin resistance. Thirdly, Cluster 2₇ showed the high LH/FSH group more clearly than the four clusters. Clusters 3₇ and 4₇ also reinforced the trend shown in the four clusters in which higher TSH seemed more associated with higher androgens rather than metabolic dysregulation.

4.3. Limitations

It must be noted that interpreting summary statistics from a multi-dimensional clustering analysis is difficult. In particular, distributions that seem to overlap in the orthogonal spaces may be quite distinct when considered across the hyperparameter space. This is particularly true in endocrine dysfunction analysis. Isolation of parts of the endocrine systems pertinent to PCOS returns a very complex system [28]. While most of the system is likely to overlap the healthy state, those with distinct phenotypes of PCOS may yield irregularity in certain subsections of the endocrine system. This is why there can appear to be little distinction between clusters when a single plot is viewed in isolation. However, while it is difficult to visually represent the clusters in multi-dimensional space, the stability of the outcomes confirms that there were stable and important distinctions between clusters.

An important limitation of all clustering is that most algorithms will identify clusters whether or not natural clusters exist. Internal validation methods like the elbow method and the stability assessment used in this paper help ensure the clusters found are the most meaningful. External validation was also undertaken by comparing the clusters obtained with those found in the analysis of Dapas et al. [14] that used an independent data-source. Future research into PCOS may involve validating the subreddit clusters with other PCOS datasets. Some limitations of k-means clustering specifically are that the number of clusters must be chosen pre-clustering and that it is sensitive to outliers [17]. We mitigated these limitations with the elbow method and by removing extreme

outliers, respectively.

While the PCOS subreddit is a new and interesting source of data, using it introduces unique data processing issues. The self-reported nature of the data may have led to less accuracy in relation to reporting the results and more outliers. Furthermore, the lack of consistency among which results a user presented led to the need for large amounts of data imputation. However, only real, non-imputed data was considered after the clustering was completed. Nonetheless, while sporadic, the dataset was sufficiently large to obtain statistically meaningful and stable results. Furthermore, trends already located in the literature were observed in this data. For example, the correlation between high androgens and high TSH [25].

Assessing stability by clustering only 90 % of the data proved to be a useful internal validation method. Using the split data method, Figs. 3–5 showed that the four clusters had high stability, while the seven clusters had relatively lower stability. However, select clusters among the seven clusters were still highly stable, such as Clusters 2₇, 5₇, 6₇ and 7₇ (Appendix C). These clusters were the most reproductively dysregulated, normal, insulin resistant, and diabetes like groups, respectively.

It must be noted that the subreddit cohort would have a different bias to typical clinical studies. In particular, the subreddit cohort only contains information from those willing and able to report their data in the public, yet anonymous, internet forum, leading to a form of selection bias. The participants are also recounting their perception of events with complete control over what details they include, leading to possible recall bias. The subreddit may also include content from some individuals who had symptoms of PCOS from a different distinct illness. However, typical cohorts in traditional trials are often affected by outpatient attendance, or healthcare system trust/accessibility, leading to their own selection biases. The frequent exclusions in clinical studies also lead to significant prevalence-incidence bias [18]. PCOS in particular seems very subject to prevalence-incidence bias due to the strong association of severe PCOS with other conditions such as diabetes [8,9] and hypothyroidism [10] and the increased likelihood of mild or lean PCOS being undiagnosed or unstudied [29–31]. The PCOS subreddit population bypasses many of the biases seen in traditional clinical trial design, including some forms of prevalence-incidence bias. Each of these two methods for studying clinical populations (traditional versus internet based), is susceptible to its own distinct set of biases [32].

4.4. Overall findings and potential

In this study, the PCOS subreddit laboratory test results dataset was successfully clustered, leading to the identification of several interesting phenotypes. This serves as further validation that data can be obtained

from internet forums, such as the PCOS subreddit, and used for clinical research. While the open contribution criteria for posting in the PCOS subreddit led to clusters that would be less populated in cohorts of formally diagnosed PCOS out-patients, the key clusters found in scientifically generated PCOS data were still clearly identifiable within the subreddit clusters.

Defining accurate phenotypes with methods such as clustering is very important for a varied condition such as PCOS. The identified groups with primarily reproductive or metabolic features of PCOS are likely to follow distinct treatment pathways. Furthermore, identifying subgroups within the indeterminate group that presents with aspects of both reproductive and metabolic features is important because treatment for the individuals within the indeterminate group could be anything from needing both types of intervention to needing entirely different types of intervention depending on other symptoms present. For example, it is possible that the higher TSH groups within the seven subreddit clusters may benefit most from optimisation of thyroid treatment rather than an insulin sensitising or traditional hormonal treatment. However, before such treatment can be trialled prospectively, there needs to be more research on PCOS clusters and how they respond to different forms of treatment. Identifying different possible groupings is an important first step in predicting treatment responses. Learning more about the subgroups within PCOS may also clarify which PCOS individuals are likely to be subject to which associated risks. For example, we may be able to distinguish whether the increased risk of diabetes associated with PCOS [8,9] does not affect the reproductive group. Such hypotheses could potentially be explored with traditional clinical trials.

The success of this study pioneers the possibility of further work with the PCOS subreddit laboratory test results dataset. Similar methods could potentially be used to extract other useful PCOS data from the PCOS subreddit. Alternatively, these methods could be used to explore other conditions using similar internet forums. The abundance of the data on such internet forums and the self-reported nature could lead to other uniquely useful datasets.

Appendix A. Data Imputation

The SciKit-Learn KNNImputer as we used it calculated the Euclidean distance between all samples while ignoring missing values. This means that the distance between two datapoints is calculated using only the features that are present in both datapoints. Equation (1) shows the calculation of Euclidean distance where ' $\vec{p}$ ' and ' $\vec{q}$ ' are vectors of equal length and ' n ' is the length of the vectors.

$$d(\vec{p}, \vec{q}) = \sqrt{(p_1 - q_1)^2 + (p_2 - q_2)^2 + \dots + (p_n - q_n)^2} \quad \text{Eq. 1a}$$

A distance matrix is created to record the distances between all datapoints as given by the missing value ignoring Euclidean distance. For each value that is missing in a datapoint, the nearest neighbours that also have a value for the feature that is missing are gathered. In this study, 7 neighbours were gathered. Since the neighbours chosen is dependent on whether the neighbour contains the missing feature, different missing values in the same datapoint may be approximated using different neighbours. The feature of interest from the gathered neighbours is averaged to produce the imputed value for the missing value in the datapoint.

Appendix B. Reference Ranges

	Reference Range	Unit	Source	Notes
TOTAL TESTOSTERONE	9–48	ng/dL	[33]	
DHEA-S	60–330	mcg/dL	[34]	
BMI	20.0–24.9	kg/m ²	[35]	
LH/FSH	0.5–2	Unitless	[36]	
FREE TESTOSTERONE	0.6–3.1	pg/mL	[37]	
FASTING GLUCOSE	70–105	mg/dL	[34]	
FSH	5–20	U/L	[34]	Excluding the unlikely case of mid-cycle peak tests as this range is phase dependent.
HbA1c	<5.7	%	[38]	
LH	5–22	U/L	[34]	Excluding the unlikely case of mid-cycle peak tests as this range is phase dependent.
TSH	0.5–2.5	mU/L	[39]	Some experts prefer using 0.35–4.5 [39].
PROLACTIN	<20	ng/mL	[34]	

(continued on next page)

5. Conclusions

The laboratory test results from the PCOS subreddit dataset were successfully clustered into groups similar to those presented in a previous clinical trial that clustered PCOS laboratory tests and genetics. When data was grouped into four clusters, distinct metabolic dysregulation and reproductive dysregulation groups were identified, alongside a group with mild forms of both dysregulations and a more normally skewed group. When data was grouped into seven clusters, the metabolic group separated into a diabetes-like and insulin resistant group. Hypothyroid and hyperandrogenic groups also emerged from the reproductive group. These groups and subgroups identified may be useful to inform future research into appropriate treatment plans for the different presentations of PCOS. Following on from previous work, this study further validates the usefulness of data extracted from the PCOS subreddit and the possibility of using other internet forums for similar clinical research.

CRedit authorship contribution statement

Rebecca H.K. Emanuel: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Data curation, Conceptualization. **Paul D. Docherty:** Writing – review & editing, Supervision, Conceptualization. **Helen Lunt:** Writing – review & editing, Supervision. **Rua Murray:** Writing – review & editing, Supervision. **Rebecca E. Campbell:** Writing – review & editing.

Declaration of competing interest

.None Declared.

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(continued)

	Reference Range	Unit	Source	Notes
OESTRADIOL	14–57	pg/mL	[34]	This range is cycle phase dependent so a range that encompasses both phases was chosen.
FASTING INSULIN	5–15	mU/L	[40]	
PROGESTERONE	3–20	ng/mL	[37]	
SHBG	32.4–128	nmol/L	[41]	This range is for progesterone during the Mid and Luteal phases. Follicular phase progesterone is typically very low.
AMH	1.26–5.6	ng/mL	[42]	
HOMA-IR	<2.5	–	[43]	
17-HYDROXYPROGESTERONE	20–500	ng/dL	[37]	This range is cycle phase dependent so a range that encompasses both phases was chosen.
TOTAL CHOLESTEROL	150–199	mg/dL	[34]	

Appendix C. Seven Clusters

The seven cluster KDE plots for the data distribution from tests for androgens, reproductive hormones, and metabolic parameters are shown in Figs. C1–C3, respectively.

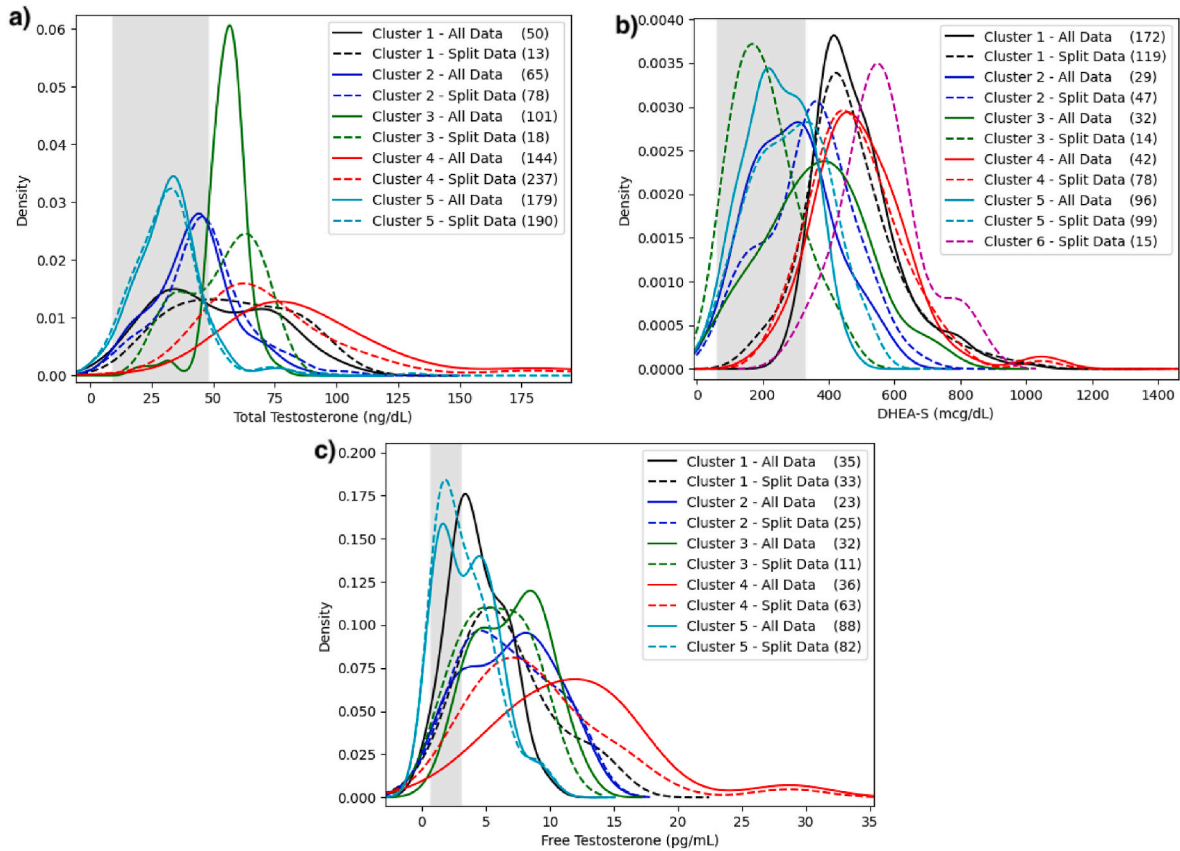


Figure C1. Kernel density estimation plots showing the distribution of androgen data across seven clusters for **a)** total testosterone, **b)** DHEA-S (dehydroepiandrosterone-sulphate) and **c)** free testosterone.

Solid lines are the clusters made using the entire dataset. Dashed lines are the clusters made when the dataset was split. Reference ranges are shaded grey. Black lines represent cluster 1, blue lines represent cluster 2, green lines represent cluster 3, red lines represent cluster 4, cyan lines represent cluster 5, magenta lines represent cluster 6, and yellow lines represent cluster 7. Groups with fewer than ten data entries are not shown. Numbers in parentheses show the number of test results for a particular cluster.

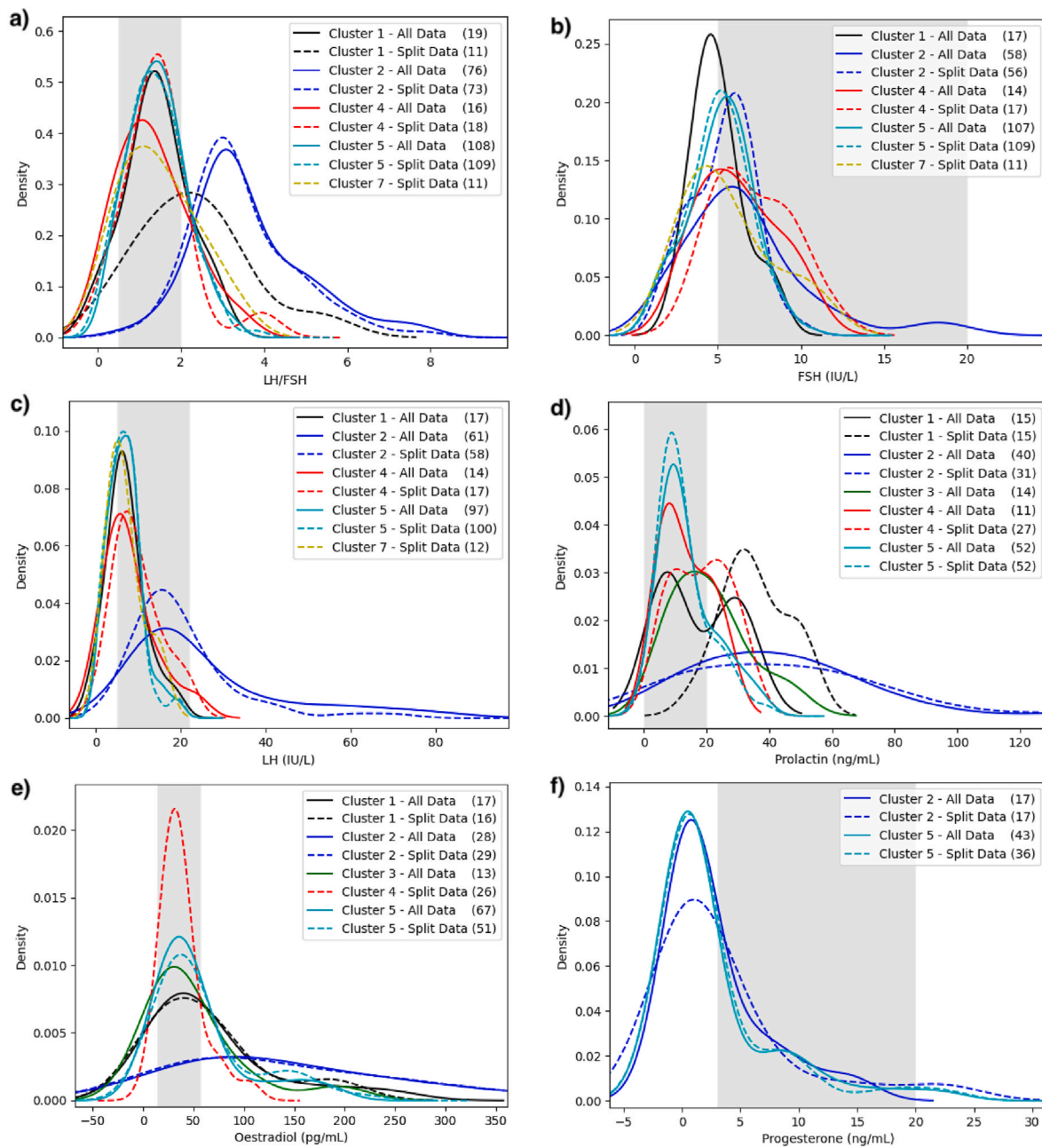


Figure C2. Kernel density estimation plots showing the distribution of reproductive hormones data across seven clusters for **a)** LH/FSH (the ratio of luteinising hormone to follicle stimulating hormone), **b)** FSH (follicle stimulating hormone), **c)** LH (luteinising hormone), **d)** prolactin, **e)** oestradiol and **f)** progesterone. Solid lines are the clusters made using the entire dataset. Dashed lines are the clusters made when the dataset was split. Reference ranges are shaded grey. Black lines represent cluster 1, blue lines represent cluster 2, green lines represent cluster 3, red lines represent cluster 4, cyan lines represent cluster 5, magenta lines represent cluster 6, and yellow lines represent cluster 7. Groups with fewer than ten data entries are not shown. Numbers in parentheses show the number of test results for a particular cluster.

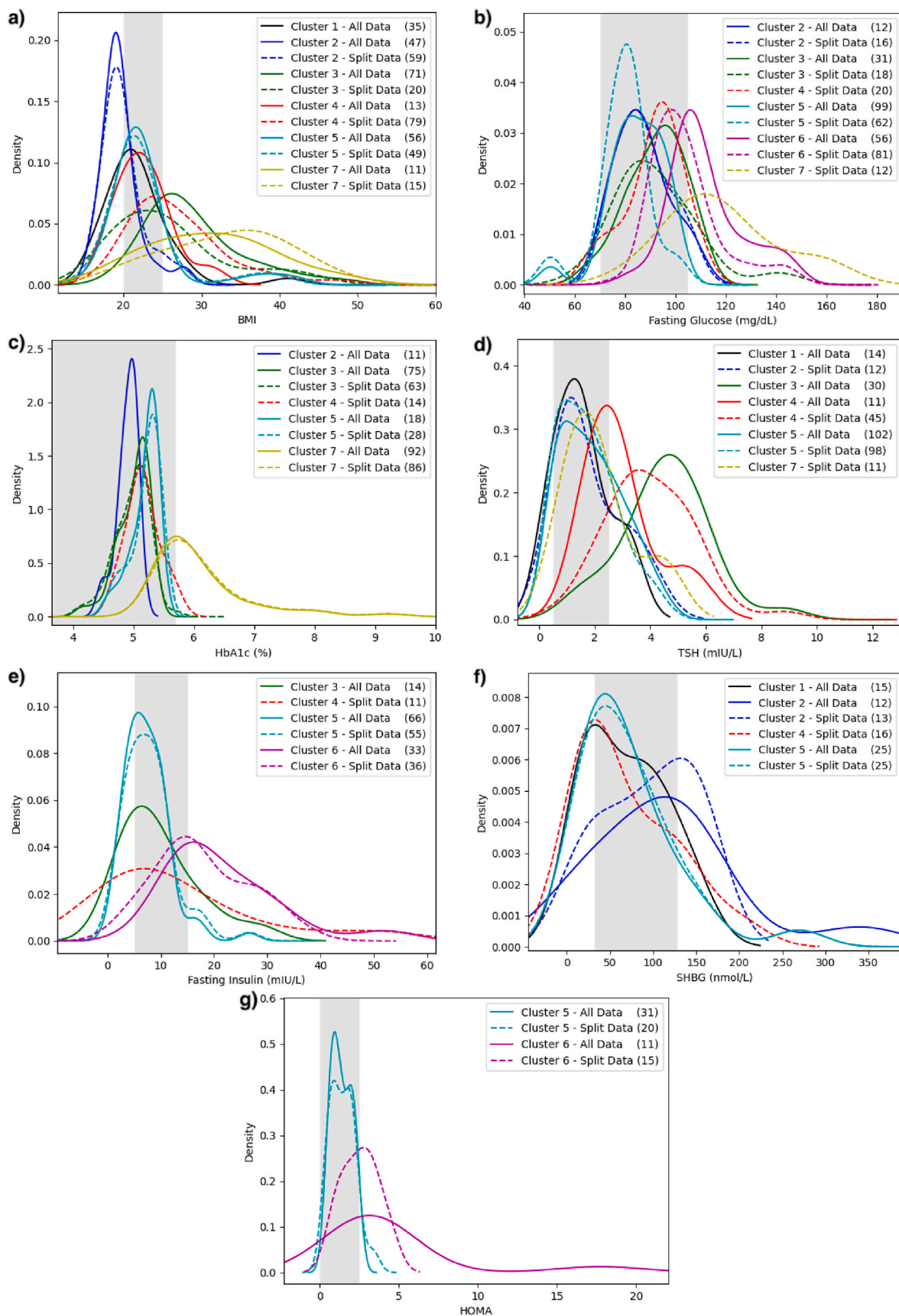


Figure C3. Kernel density estimation plots showing the distribution of metabolic test data across seven clusters for **a)** BMI (body mass index), **b)** fasting glucose, **c)** HbA1c (glycated haemoglobin), **d)** TSH (thyroid stimulating hormone), **e)** fasting insulin, **f)** SHBG (sex hormone binding globulin), **g)** HOMA-IR (homeostasis model assessment-estimated insulin resistance).

Solid lines are the clusters made using the entire dataset. Dashed lines are the clusters made when the dataset was split. Reference ranges are shaded grey. Black lines represent cluster 1, blue lines represent cluster 2, green lines represent cluster 3, red lines represent cluster 4, cyan lines represent cluster 5, magenta lines represent cluster 6, and yellow lines represent cluster 7. Groups with fewer than ten data entries are not shown. Numbers in parentheses show the number of test results for a particular cluster.

Cluster 1₇ was characterised by mildly high androgens (Fig. C1). Cluster 2₇ had higher LH/FSH (Fig. C2a) and prolactin (Fig. C2d) while seemingly the most metabolically healthy with fasting glucose (Fig. C3b), HbA1c (Fig. C3c) and higher SHBG (Fig. C3f). Individuals in this cluster also had the most normally skewed oestradiol (Fig. C2e) and underweight BMI (Fig. C3a). Cluster 3₇ was another group with mildly high androgens (Fig. C1), but it was also characterised by a mildly high BMI (Fig. C3a) and the highest TSH (Fig. C3d). Cluster 4₇ was a distinctly hyperandrogenic group with the highest total testosterone (Fig. C1a) and free testosterone (Fig. C1c). Similar to Cluster 3₇, Cluster 4₇ also skewed higher for BMI (Fig. C3a) and TSH (Fig. C3d).

Cluster 5₇ seemed to be the most normally skewed group among the seven clusters, with low androgens (Fig. C1) and a lack of strong reproductive (Fig. C2) or metabolic (Fig. C3) dysregulation. Cluster 6₇ had strong metabolic dysregulation with high fasting glucose (Fig. C3b), fasting insulin (Fig. C3e) and HOMA-IR (Fig. C3g). It also appeared to have the highest DHEA-S (Fig. C1b), which could indicate Cluster 6₇ contained individuals with the most overactive adrenal cortex androgen production [44]. Cluster 7₇ appeared to be the most obese group with the highest mean BMI (Fig. C3a). It also appeared to be the group with the most T2DM and prediabetes, based on high HbA1c (Fig. C3c) and fasting glucose levels (Fig. C3b).

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